3D Printing

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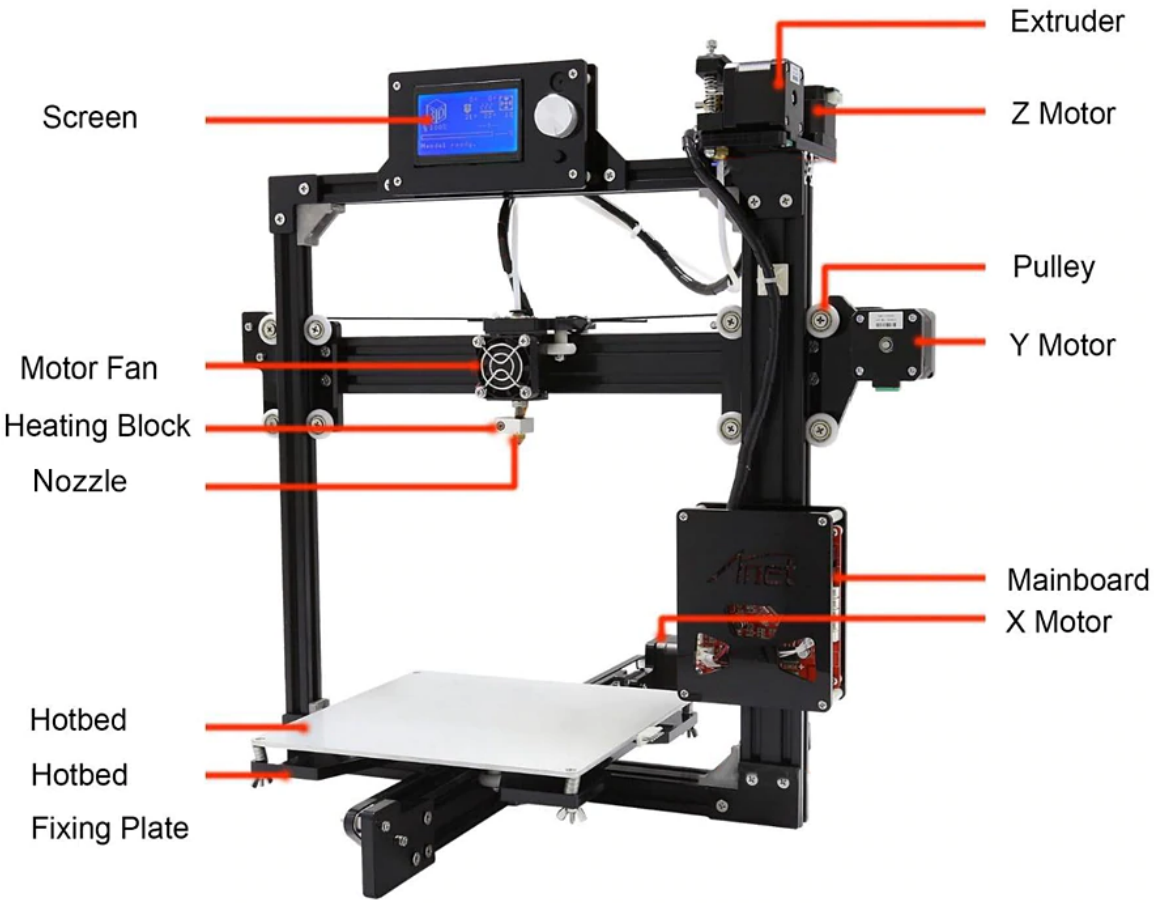
3D Printer parts

- extruder : estrusore
- spool : rotolo
- nozzle : ugello
- print bed : piatto

Components Name



Phases

Modelling	Slicer
Autodesk Fusion 360	Cura
Autodesk 123	Simplify 3D
Onshape	Slicer
Blender	Idea Maker
Zibrush ($)	Kisslicer
Sculptrist	[CraftWare]
Sketch up	
Online database	Pronterface
Thingiverse	Repetier - Host

Notes:

- STL : Mesh polygonal. The item is made of many triangles

Materials

Name	Usage	Temperature	Speed	Cost	Notes
PLA	Generic	Lower		Cheapest	Easiest to use
PET		Same PLA	Slower		
ABS					derived from petrol Toxic More robust
PETG					
Nylon	Mechanic piece	High			

> https://www.prusa3d.com/material-guides/

Links

- RepRap
- stampa3d-forum

Make gcode

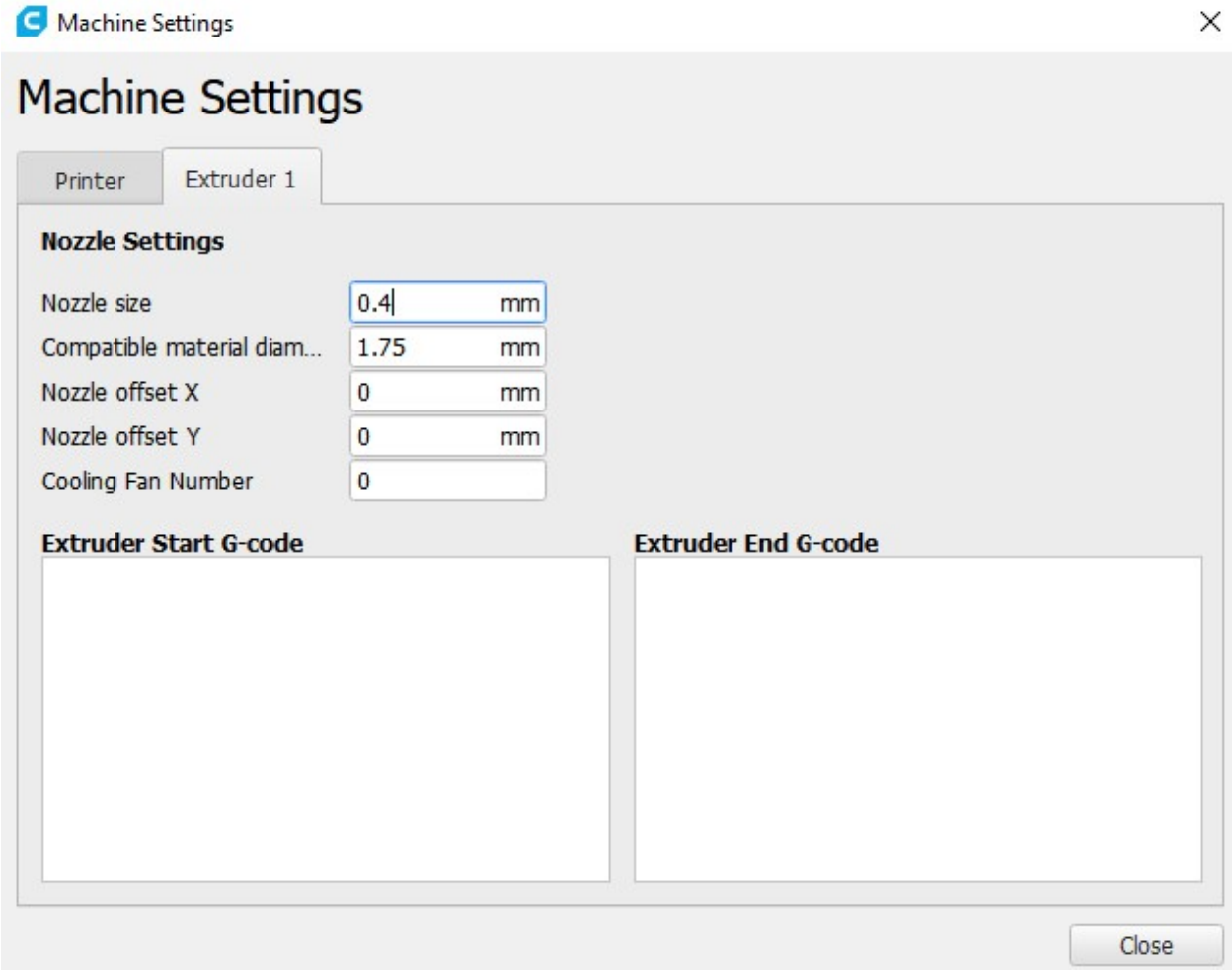
Cura

1. Open stl file with Cura or any other slicer program.
2. Select your printer (if not present, create a custom profile and set manufactures values) see here
3. Adjust your item if required
4. Set all printing settings
5. Click on `Prepare` - Waiting for the slicing
6. Click on `Save to File` in order to save the gcode file

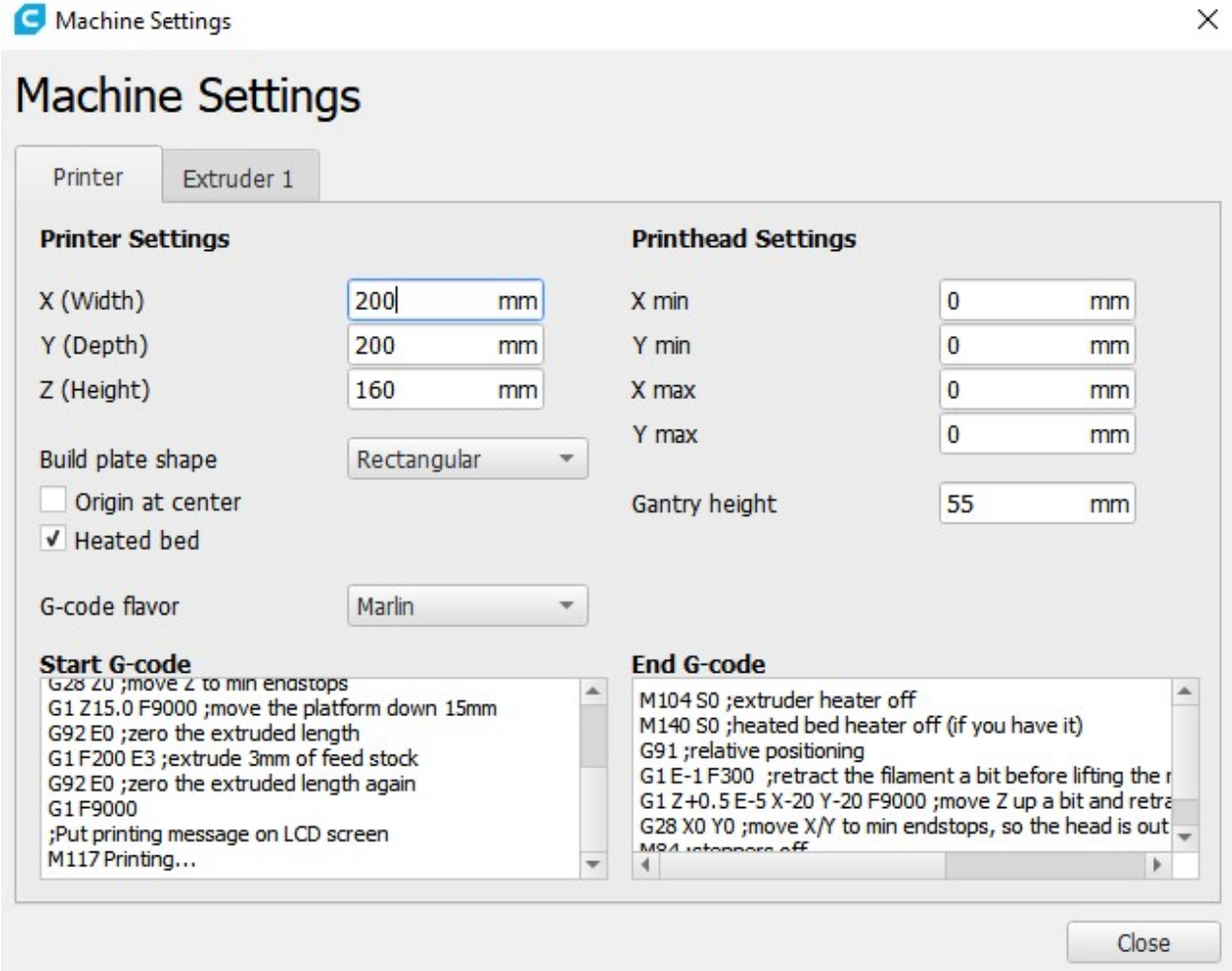
Profiles

Prusa i3

- **Machine Settings - Extruder**



- **Machine Settings - Printer**



Printing Steps

1. Calibrate the print bad
    1. `Prepare` --> `Auto Home` : reset axis
    2. `Prepare` --> `Disable steppers` : disable steppers in order to move them during the calibration
    3. Move the nozzle and the print bad to the 4 corners and the centre. For each position (5 in total), take a A4 paper a move it along the surface: make sure the extruder does not obstruct the movement and the friction is always the same

    > In case of adjusting, use the screws in the 4 corners

    https://help3d.it/come-calibrare-il-piano-di-stampa/

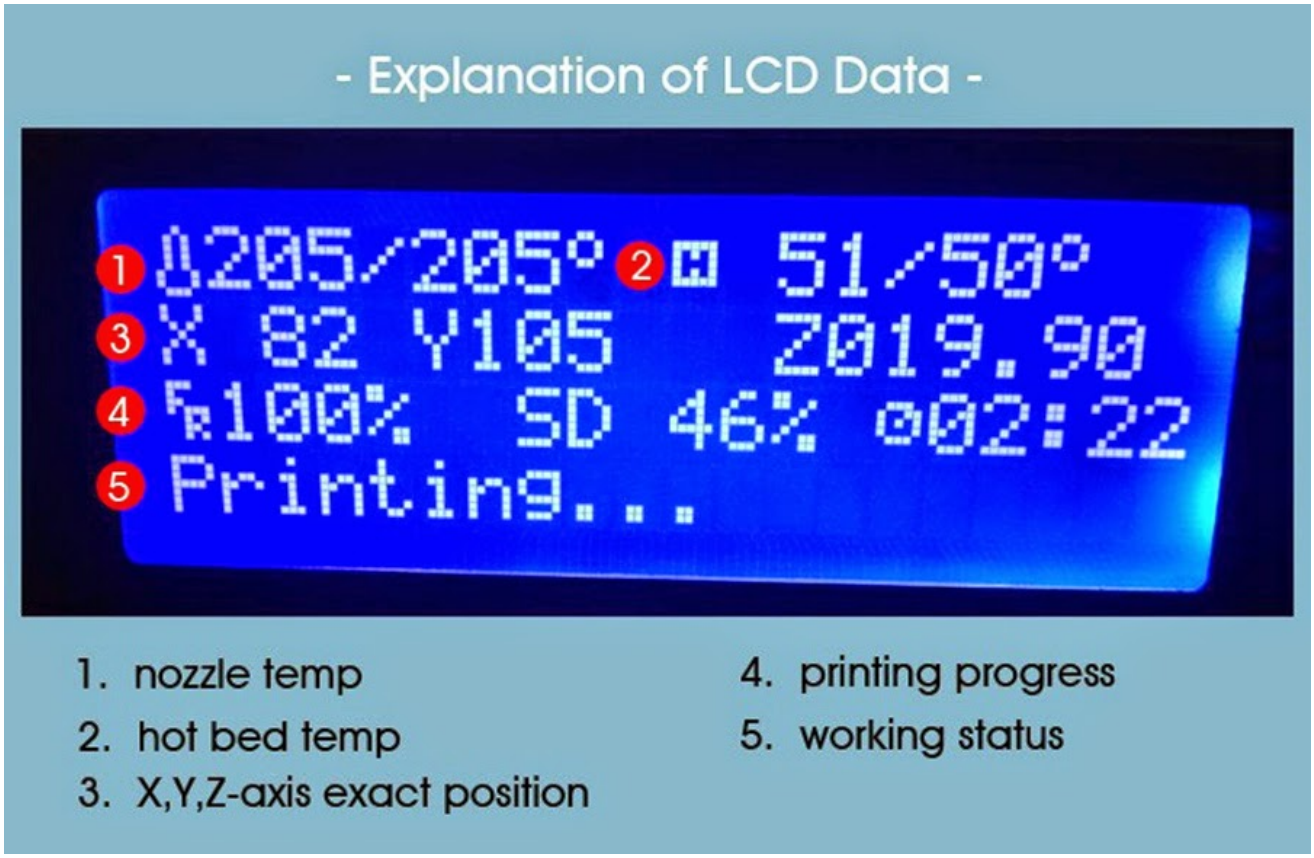
2. Insert the filament:

    press on the spring and insert the filament into the nuzzle. Release the spring

3. On the printer controller: `Prepare` --> `Preheat PLA` --> 0.2 mm (Profile)
4. Remove the residual material from the nozzle:
    1. `Prepare` --> `Move axis` --> Select `Move 1mm` (or `Move 0.1 mm`) --> `Extruder` --> Rotate in order to get the material getting out

> The Extruder must be heat in order to making this point working

5. Clean the surface making sure is not warm. Use alcohol
6. Select the the object to print:
    1. `Print from SD` --> Select the gcode file to print

> During the printing, you can change the extruder temperature

Display info